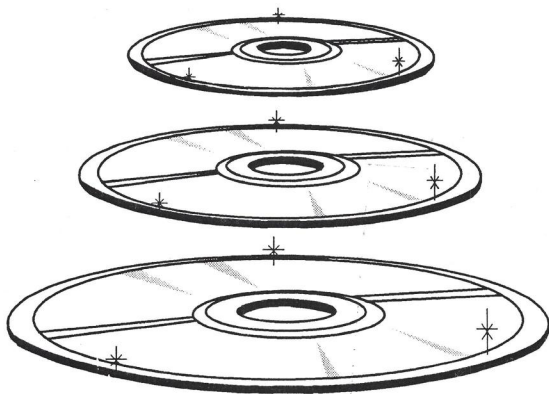


Troubleshooting Guide

Compact Disc Players
1986 to 1990 Model Years



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Introduction

The widespread popularity of Hi-Fi Digital Audio has made the compact disc player a common place component in today's audio systems. Fortunately, in the years since its initial introduction, the serviceability of the compact disc player has improved dramatically, mostly due to advances in digital technology. What this technology was not capable of doing however, is to do away with the various mechanical and electrical adjustments required on these units.

This guide is meant to eliminate trial-and-error servicing procedures and aid in the repairing of compact disc players by standardizing the procedure. If followed correctly, the optimum operating condition of the unit can be achieved by using this guide. For this reason, it is important that all of the procedures listed in this guide be performed BEFORE replacing any optical pickup assemblies.

Included along with this procedure is a useful reference chart containing pickup part numbers and related information.

**NOTE: THE FOLLOWING PROCEDURES CONTAIN GENERAL INFORMATION.
FOR SPECIFIC DETAILS PLEASE REFER TO THE CORRESPONDING
SERVICE MANUAL FOR YOUR UNIT.**

Models

All compact disc players from 1986 to 1990.

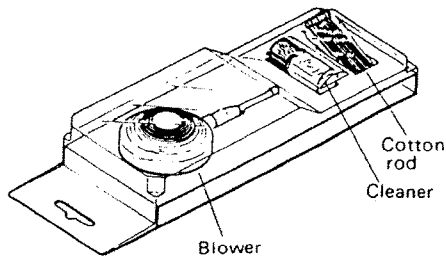
Possible Symptoms

1. Skipping.
2. TOC is read, but disc does not play.
3. No TOC.
4. Poor playability.

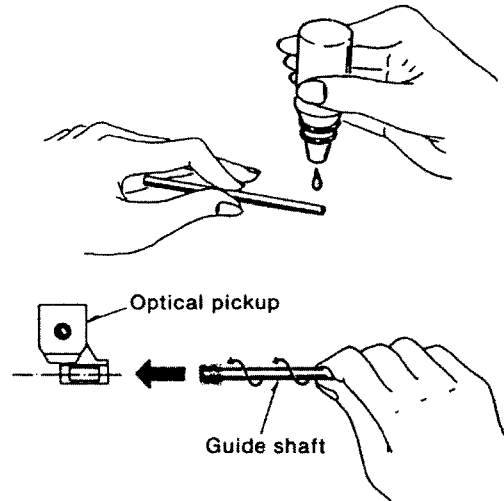
Procedure

Step 1. Clean the lens and guide rails

Use the cleaning (Part Number SZZP1038C) to clean the lens. Moisten a cotton rod with cleaning solution and wipe the optic lens from center to edge. Clean the guide rails by using a lint-free cloth to remove any visible dirt or dust. Lubricate the guide rails by applying the prescribed traverse oil (see page 8).

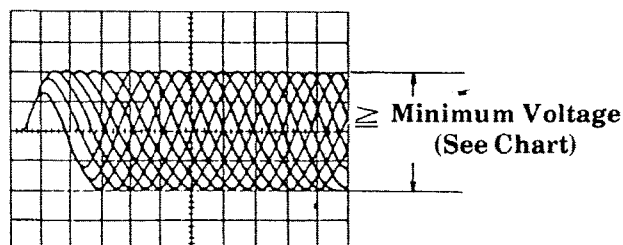


Lens cleaning kit (Part No. : SZZP1038C)



Step 2. Observe RF waveform

Connect an oscilloscope to the RF test point (refer to S/M for specific location) and play track 8 of the playability test disc CDT-016 (Part Number SZZP1054C). Observe the waveform and compare it to the minimum peak to peak voltage allowed for it (see page 8). If the observed waveform is low, it is first necessary to perform the next step BEFORE replacing the optical pickup unit.

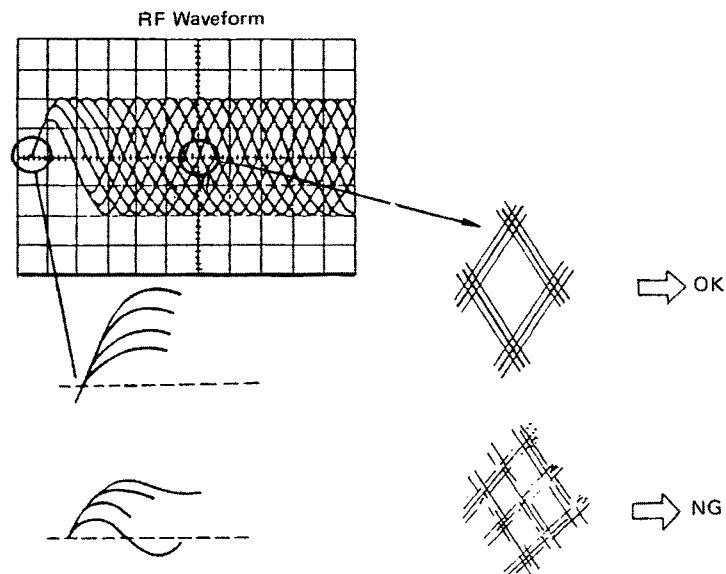


RF Waveform

Step 3. Improve RF level

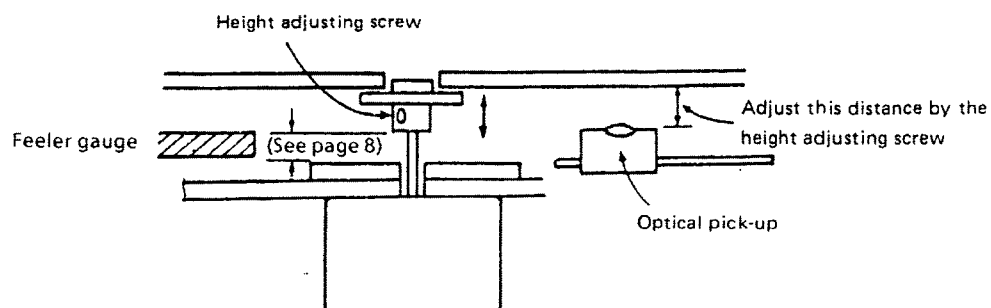
A. Perform best eye adjustment

While still observing the RF waveform, rotate the best eye adjustment pot to obtain the largest RF amplitude while achieving of a clear eye pattern. This adjustment sets the focus servo's operating reference voltage.



B. Adjust turntable height

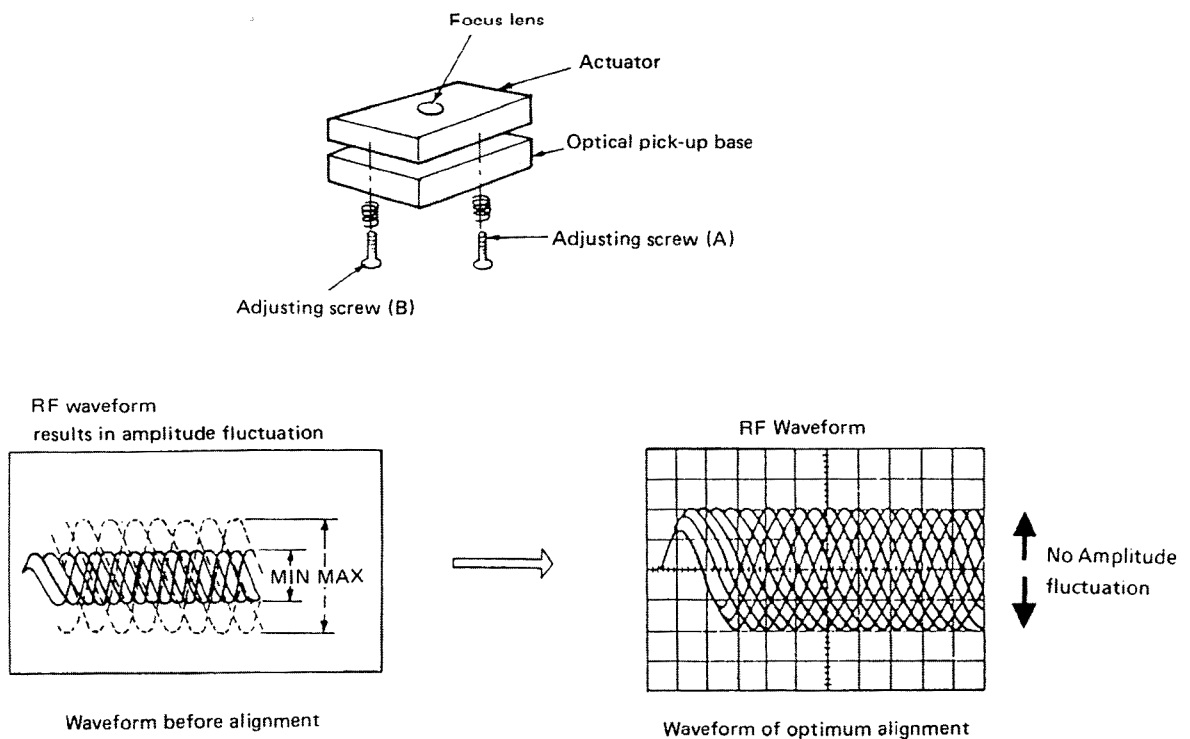
The disc may be beyond the operating range of the focus servo. If so, it is necessary to position the turntable to the physical midpoint of the servo's operating range. Play track 1 of the playability test disc and connect the oscilloscope to the focus offset test point. Adjust the turntable height so as to approach 0 volts as close as possible. This adjustment is a must when mechanical parts in the traverse deck are replaced or removed.



Step 3. Improve RF level (continued)

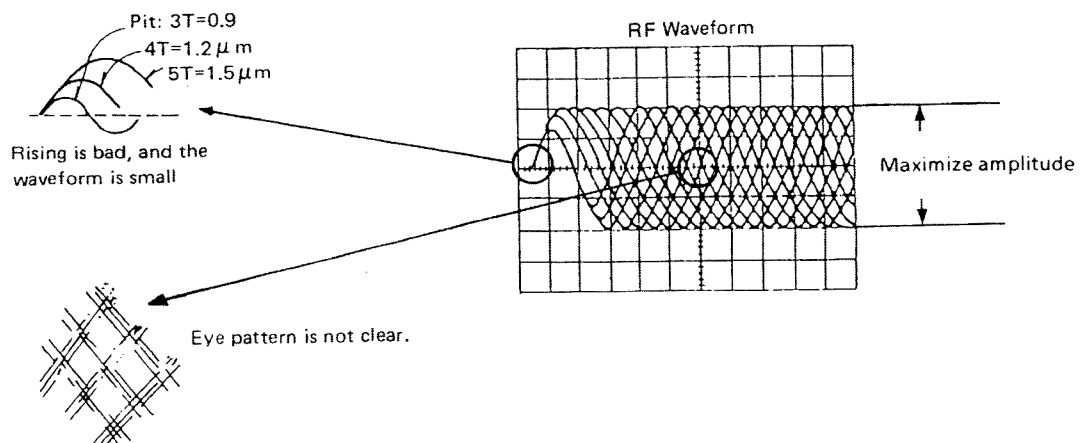
C. Mechanical adjustment of optical pickup

The purpose is to make the base of the optical pickup perpendicular to the test disc. To simulate a worst - case condition, the uneven test disc CDT-018 (Part Number SZZP1056C) is used. Play track 8 of the test disc and observe the RF waveform. The mechanical adjustment screws should be adjusted so that the amplitude fluctuation of the RF is minimized.



D. Re-adjust for best eye

Turntable height and mechanical adjustments affects the best eye, therefore it should be readjusted.



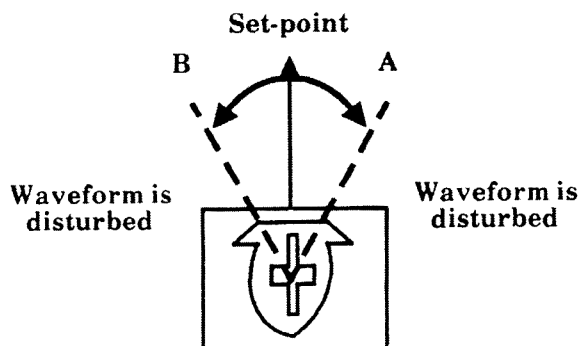
Step 4. Improve playability

A. Test focus and tracking servo circuits

Playing the 0.9mm wedge (track 8) on the playability test disc serves to show how well the unit is at tracking defects on discs. If skipping occurs, first inspect the test disc for scratches or dust. If skipping persists, it may be necessary to perform the mechanical adjustment over again (See Step 3C).

B. Perform PLL adjustment

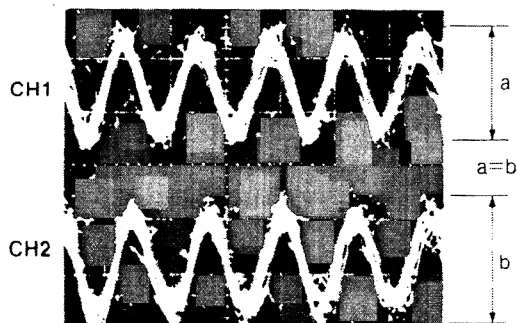
Play the 0.8mm wedge on the playability test disc and connect the oscilloscope to the PLL test point. Adjust the PLL variable resistor so its position is between the two locations where the PLL waveform starts to be disturbed. This adjustment is used to set the operating range of the voltage controlled oscillator that allows the disc to be synchronized with the mechanism's internal clock.



Step 5. Electrical adjustments

A. Focus gain adjustment

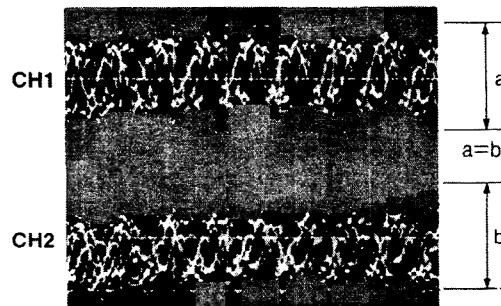
Connect the servo gain adjuster (Part Number SZZP1017 or SZZP1094C) as explained in the Service Manual for the model being serviced. Connect channels 1 and 2 of the oscilloscope to TP1 and TP2 on the servo gain adjuster. Play track 1 of the playability test disc and turn the focus gain adjustment pot so that the wave forms at TP1 and TP2 are of equal amplitude. This is done to set the focus gain to unity, thereby stabilizing the lens focus.



Step 5. Electrical adjustments (continued)

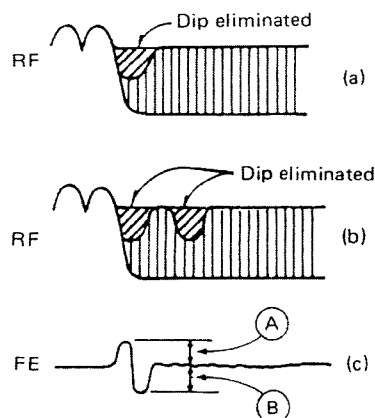
B. Tracking gain adjustment

Change the mode select switch on the servo gain adjuster to position "2" and select the correct oscillator frequency. Adjust the tracking gain variable resistor so that the amplitudes at TP1 and TP2 are equal while playing track 8 of the playability test disc. This adjustment makes the gain of the tracking servo equal to 1.



C. Focus offset adjustment

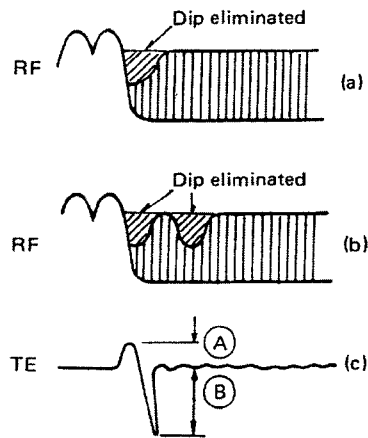
Connect channels 1 and 2 of the oscilloscope to the RF and TE test points on the CD player's PCB. Playback the black band test disc (Part Number SZZP1057C). This will simulate manufacturing defects on CDs and looks like a "dip" on the RF waveform. Adjust the focus offset pot so that the dip is minimized or eliminated. The FE waveform must be adjusted so that A and B are at minimum or balanced.



Step 5. Electrical adjustments (continued)

D. Tracking offset adjustment

Perform the same procedure as for focus offset except use the TE testpoint. This adjustment allows the tracking servo to compensate for horizontal control of the focus servo. Adjust the tracking offset pot so that the dip(s) is (are) minimized or eliminated. The TE waveform must be adjusted so that A and B are balanced.



MODEL NUMBERS	PICK UP	MIN. RF	LUBRICANT	T/T HEIGHT		
SL-NP3, XP7 RX-CD70, FD80, CD8936	SOAD10A	0.84V	SZZ0L24	N/A		
SL-P100, P105, P106, P300, P500, P3620, PJ11, PJ33, CD8956	SOAD20A	0.77V	SZZ0L32			
SL-P111,P115,P116,P117,P118, P110,P210,P310,P420,P520, P720,P1000,P3500,P3600,P3700, PJ20,CD8946,CD8947,CS47U	SOAD30A	0.84V	RZZ0L05	0.9mm		
SL-P217, P218, P220, P320, P120, RX-CD100	SOAD32A			N/A		
SL-P600C, P3900C	SOAD32A-2			0.9mm		
SL-P800C	OPTIMA-20	-		0.5mm		
SL-XP5,NP10,CD8966 RX-FD55,FD75,FD80	SOAD40A	0.91V	SZZ0L24	N/A		
SL-P1200,PL1200X P1300	SOALP1200	0.84V	RZZ0L05			
SL-P350,P550,P770,P990	SOAD60A	1.0V	SZZ0L31	0.9mm		
SL-P127,P128,P150,P3510, P200,P230,P250,P555,P999. PS352,P170,P370,P101,P102, P222, CD8930,CD8948,CD8949	SOAD70A	0.39V				
SL-P400C,P405C,P406C,P3800C, CD8988						
SL-PC10,PC20,PC30, PC11,PC14, PC25,PC33,PC15,PC45,PC362, PC363,CD8989.						
SL-XP6,NP11	SOAD100A	0.45V	SZZ0L30	N/A		
RX-DS30,DS650	SOAD101A	0.39V	SZZ0L31			
RX-DT5,DS20 DS40,DS660	SODD101-3Z		N/A			

N/A - NOT APPLICABLE